Edwin Aguirre

edwina3@uci.edu | edwin-aguirre.itch.io | github.com/Edwin-Aguirre

SKILLS

Languages: C#, GDScript, C++
Engines & Tools: Unity, Godot, UE5, Photon PUN/Fusion, Git
Specialties: Multiplayer Networking, AI (FSM/BT), Physics, Optimization, Rapid Prototyping

PROJECTS

FARMING TERRITORY: REGROWN

Gameplay Programmer | Godot, GDScript, AI Systems | Shipped on Itch.io | 2025

- Engineered a **modular resource management system** in GDScript using the **Observer Pattern**, allowing for dynamic territory expansion and real-time UI updates for 10+ resource types.
- Authored **non-deterministic AI Behavior Trees** with weighted decision-making logic, enabling agents to execute complex tactical pivots based on player expansion strategies.
- Implemented a **data-driven economic engine** using Godot Resources, decoupling game balance variables from core logic to allow for rapid iteration and "hot-swapping" of unit stats.

MIMIC

Lead Gameplay Programmer | Unity, C#, Git | Capstone Project |2024

- Directed the technical architecture of an **extensible Player Controller** and multi-layered interaction system, utilizing **C# Interfaces** to handle physics-based movement and context-sensitive world objects.
- Programmed a modular **Finite State Machine (FSM)** for enemy AI, integrating **Unity NavMesh** for pathfinding and dynamic obstacle avoidance in complex 3D environments.
- Established a **reusable component library** and a centralized **C# Event Bus**, reducing feature implementation time for the design team by **~30%** over the 6-month development cycle.

WATERFIGHT!

Gameplay Programmer | Unity, C#, Photon Unity Networking | 2022

- Developed a **real-time synchronization layer** using Photon PUN, optimizing state replication and **Remote Procedure Calls (RPCs)** to maintain high-fidelity game state across 6-player sessions.
- Architected **network-safe gameplay systems** featuring **client-side prediction** and server reconciliation, effectively masking up to **150ms of latency** for fast-paced balloon combat.
- Conducted rigorous **network performance profiling** and packet analysis to minimize overhead, ensuring a consistent **60 FPS** experience during peak gameplay.

EDUCATION

University of California, Irvine
B.S. Game Design and Interactive Media - Cum Laude
GPA: 3.9 | Donald Bren School of Information and Computer Sciences
Focus: Gameplay Systems, Multiplayer Networking, and AI Behaviors